

Compact Stars in DRLT: Neutron Stars, Quark Stars, and the $\mathbb{C}^3$ Confinement Safety Valve

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Abstract

We derive the structure and stability of compact stars entirely within the DRLT framework, using only $d = 5$ lattice geometry and zero free parameters. The key new ingredient is that the Planck constant $\hbar_{\text{eff}} \propto \sqrt{\det(G_h)}$ is dynamical: it decreases under compression as ψ vectors align. This introduces a density-dependent correction to the standard degeneracy pressure: $P_{\text{deg}}^{\text{DRLT}} = \det(G_h) \times P_{\text{deg}}^{\text{standard}}$. We prove that (i) neutron stars are stabilized by $\mathbb{C}^3$ confinement, which sets a minimum $\det(G_h)$; (ii) pure quark stars are unstable due to a positive feedback loop triggered by $\mathbb{C}^3$ deconfinement; (iii) the viscosity of the quark-gluon core follows $\eta/s = 1/(4\pi)$ (KSS bound) from $c \times 2\pi$ where $c = 2$; and (iv) quark cores can exist only inside neutron star shells above $\sim 2 M_\odot$. These predictions are testable via gravitational wave observations of neutron star mergers.

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1 DRLT Degeneracy Pressure

1.1 Standard vs. DRLT formulation

In standard quantum mechanics, the degeneracy pressure of a non-relativistic Fermi gas is:

$$P_{\text{deg}}^{\text{std}} = \frac{(3\pi^2)^{2/3}}{5} \cdot \frac{\hbar^2 n^{5/3}}{m} \quad (1)$$

where $\hbar = 1.054 \times 10^{-34} \text{ J} \cdot \text{s}$ is a universal constant, n is the number density, and m is the particle mass.

In DRLT, $\hbar$ is not a constant but a local dynamical field (Chapter 4 of the main text):

$$\hbar_h = \frac{\sqrt{\det(G_h)}}{4 \ln 2} = \frac{A_h}{4 \ln 2} \quad (2)$$

where $\det(G_h)$ is the Gram determinant of the local hinge (triangle) formed by three neighboring vertices.

Substituting $\hbar \rightarrow \hbar_{\text{eff}}$ in Eq. (1):

$$P_{\text{deg}}^{\text{DRLT}} = \frac{\det(G_h)}{16 \ln^2 2} \cdot \frac{(3\pi^2)^{2/3}}{5} \cdot \frac{n^{5/3}}{m} \quad (3)$$

The DRLT correction factor is:

$$\frac{P_{\text{deg}}^{\text{DRLT}}}{P_{\text{deg}}^{\text{std}}} = \frac{\det(G_h)}{\det_0} \quad (4)$$

where $\det_0 \approx 0.49$ is the Gram determinant for random (dilute) matter. In standard physics, this ratio is implicitly set to 1; DRLT makes it dynamical.

1.2 Density dependence of $\det(G_h)$

As matter is compressed, neighboring ψ vectors are forced to align (their overlap $|G_{ij}|$ increases toward 1). This drives $\det(G_h) \rightarrow 0$:

$$\det(G_h)(\rho) = \det_0 \cdot \exp\left(-\frac{\rho}{\rho_{\text{sat}}}\right) \quad (5)$$

where $\rho_{\text{sat}} = n_S \cdot \rho_0 = 3\rho_0$ is the $\mathbb{C}^3$ saturation density (the density at which all three spatial degrees of freedom begin to overlap), and $\rho_0 = 2.3 \times 10^{17} \text{ kg/m}^3$ is nuclear saturation density.

The exponential form follows from the Wishart eigenvalue distribution of the Gram matrix: each independent compression mode contributes a multiplicative factor $e^{-\Delta\rho/\rho_{\text{sat}}}$.

ρ/ρ_0	$\det(G_h)$	$\hbar_{\text{eff}}/\hbar_0$	Physical state
0.5	0.41	0.92	Normal nuclear matter
1.0	0.35	0.85	Nuclear saturation
3.0	0.18	0.61	$\mathbb{C}^3$ saturation onset
5.0	0.09	0.43	Partial deconfinement
10.0	0.02	0.19	Deep deconfinement
20.0	0.001	0.04	Near-collapse

1.3 The competition: $n^{5/3}$ vs. $\det(G_h)$

In standard physics, P_{deg} grows monotonically with density ($\propto n^{5/3}$). In DRLT, two effects compete:

- **Fermi statistics:** $n^{5/3}$ increases with compression. (Stabilizing.)
- **ψ alignment:** $\det(G_h)$ decreases with compression. (Destabilizing.)

The net pressure:

$$P_{\text{net}} \propto \det_0 e^{-\rho/(3\rho_0)} \times \left(\frac{\rho}{\rho_0}\right)^{5/3} \quad (6)$$

Differentiating and setting $dP/d\rho = 0$ gives the maximum pressure at:

$$\rho_{\text{max}} = 5\rho_0 \quad (7)$$

Beyond this density, **degeneracy pressure decreases with increasing density**. This is impossible in standard physics but inevitable in DRLT. It sets an absolute upper bound on compact star density.

2 Neutron Star Structure

2.1 Why neutron stars are stable: the $\mathbb{C}^3$ safety valve

Neutron matter consists of confined baryons. In DRLT, confinement means quarks are restricted to $\mathbb{C}^3 \subset \mathbb{C}^5$ (Chapter 5, Sec. 5.4: $N_{\text{eff}} = 1$ for the strong force).

This confinement imposes a **minimum** on $\det(G_h)$. The $\mathbb{C}^3$ boundary prevents complete ψ alignment: even at high density, the three spatial directions maintain some independence because the temporal sector $\mathbb{C}^2$ is decoupled. Formally:

$$\det(G_h) \geq \det_{\min}(\mathbb{C}^3) > 0 \quad (\text{confinement active}) \quad (8)$$

This minimum ensures $\hbar_{\text{eff}} > 0$, hence $P_{\text{deg}} > 0$, providing a floor for the degeneracy pressure. The neutron star is stabilized by the *algebraic boundary* between $\mathbb{C}^3$ and $\mathbb{C}^2$.

2.2 Neutron star internal structure

$$\text{Neutron star} = \begin{cases} \textbf{Crust:} & \rho < \rho_0, \quad \det \sim 0.4, \quad \text{normal nuclei} \\ \textbf{Outer core:} & \rho_0 < \rho < 3\rho_0, \quad \det \sim 0.2\text{--}0.35, \quad \text{neutron fluid} \\ \textbf{Inner core:} & 3\rho_0 < \rho < 5\rho_0, \quad \det \sim 0.05\text{--}0.18, \quad \mathbb{C}^3 \text{ saturating} \\ \textbf{Center:} & \rho \sim 5\text{--}7\rho_0, \quad \det \sim 0.05, \quad \text{possible quark core} \end{cases} \quad (9)$$

The maximum mass is reached when the central density approaches $\rho_{\text{max}} \approx 5\rho_0$:

$$M_{\text{max}} \approx 2.0\text{--}2.3 M_{\odot} \quad (10)$$

consistent with the observed neutron star mass distribution (heaviest confirmed: PSR J0740+6620 at $2.08 \pm 0.07 M_{\odot}$).

3 Quark Star Instability

3.1 Deconfinement: $\mathbb{C}^3 \rightarrow \mathbb{C}^5$

At sufficiently high temperature or density, the $\mathbb{C}^3$ confinement boundary dissolves. Quarks access the full $\mathbb{C}^5$ simplex. In DRLT, this means:

- All $d^2 = 25$ Binet–Cauchy channels activate (vs. 1 channel in confinement).
- N_{eff} changes from 1 (confined) to large (deconfined).
- The $\mathbb{C}^3$ floor on $\det(G_h)$ is **removed**.

3.2 The positive feedback catastrophe

Without the $\mathbb{C}^3$ floor, the following positive feedback loop activates:

$$\text{Deconfinement} \rightarrow \det \downarrow \rightarrow \hbar \downarrow \rightarrow \text{quantum fluctuations} \downarrow \rightarrow \psi \text{ aligns more} \rightarrow \det \downarrow \downarrow \rightarrow \dots \rightarrow \det \rightarrow 0^+ \rightarrow \dots \quad (11)$$

Theorem 3.1 (Quark Star Instability). *A self-gravitating sphere of deconfined quark matter (pure quark star) is unstable in DRLT.*

Proof. Define $f(\rho) = P_{\text{deg}} - P_{\text{grav}}$ as the net pressure.

For confined matter ($\mathbb{C}^3$ active): $\det(G_h) \geq \det_{\text{min}} > 0$, hence $\hbar_{\text{eff}} \geq \hbar_{\text{min}} > 0$, hence $P_{\text{deg}} > 0$ at all densities. Stable equilibrium exists (TOV solution).

For deconfined matter ($\mathbb{C}^5$ open): $\det(G_h)$ has no lower bound. The feedback $\det \downarrow \rightarrow \hbar \downarrow \rightarrow \det \downarrow \downarrow$ gives:

$$\left. \frac{d(\det)}{d\rho} \right|_{\text{deconfined}} < \left. \frac{d(\det)}{d\rho} \right|_{\text{confined}} \quad (12)$$

The rate of $\det$ decrease accelerates. At some critical density ρ_c , P_{deg} drops below P_{grav} and no equilibrium exists. The system collapses to a black hole ($\det \rightarrow 0^+$, $\hbar \rightarrow 0^+$). $\square$

3.3 Quantitative comparison

At the same total density ρ , comparing confined (neutron) vs. deconfined (quark) states:

$$\text{Number density: } n_q = 3 n_n \quad (3 \text{ quarks per neutron}), \quad (13)$$

$$\text{Mass: } m_q \approx m_n/3 \quad (\text{constituent quark}), \quad (14)$$

$$\text{Fermi factor: } \frac{P_q^{\text{Fermi}}}{P_n^{\text{Fermi}}} = 3^{8/3} \approx 18.7. \quad (15)$$

The quark Fermi pressure is ~ 19 times larger. However, the $\det$ ratio:

$$\frac{\det_q}{\det_n} = e^{-1} \approx 0.37 \quad (\text{deconfinement penalty}) \quad (16)$$

Net pressure ratio:

$$\frac{P_q}{P_n} = e^{-1} \times 3^{8/3} \approx 6.9 \quad (17)$$

Quark degeneracy pressure is $\sim 7\times$ larger than neutron degeneracy pressure at the same density. The instability is *not* due to insufficient pressure. It is due to the removal of the $\det$ floor, triggering the positive feedback loop.

3.4 Why the pressure advantage doesn't help

The factor 6.9 is static. The feedback loop is dynamic:

Step	$\det(G_h)$	$\hbar_{\text{eff}}/\hbar_0$	$P_{\text{deg}}/P_{\text{initial}}$
Initial (deconf.)	0.09	0.43	1.00
Feedback 1	0.05	0.32	0.56
Feedback 2	0.02	0.20	0.22
Feedback 3	0.005	0.10	0.06
Feedback n	$\rightarrow 0^+$	$\rightarrow 0^+$	$\rightarrow 0$

The initial pressure advantage ($6.9\times$) is consumed in ~ 2 – 3 feedback cycles. After that, collapse is inevitable.

4 The Quark Core: Hybrid Stars

4.1 Stable configuration: quark core + neutron shell

A quark core can exist if it is **enclosed by a neutron matter shell** that provides the missing det floor externally:

$$\text{Hybrid star} = \underbrace{\text{Neutron shell}}_{\text{det} > \text{det}_{\min}, P_{\text{deg}} > 0} + \underbrace{\text{Quark core}}_{\text{det} \rightarrow 0^+, \eta/s = 1/(4\pi)} \quad (18)$$

The shell provides gravitational confinement from outside. The quark core is a region of $\mathbb{C}^5$ -active matter with $\eta/s = 1/(4\pi)$ (a perfect fluid core inside a viscous neutron shell).

4.2 Mass threshold

The quark core forms when the central density exceeds $\rho_{\text{deconf}} \approx 5\rho_0$, which occurs at:

$$M_{\text{hybrid}} \gtrsim 2.0 M_{\odot} \quad (19)$$

Below this mass, the central density never reaches deconfinement, and the star is a pure neutron star. Above it, a quark core grows with mass, until the shell can no longer support it.

5 The sQGP Phase: $\eta/s = 1/(4\pi)$

5.1 Derivation from DRLT

In the deconfined region (quark core or RHIC plasma), all $d^2 = 25$ Binet–Cauchy channels are active, but $\text{rank}(G) \leq 5$ remains absolute.

Entropy: each hinge carries 1 bit (Holevo bound), so the entropy per unit hinge area:

$$s = \frac{S}{A} = A_h \quad (\text{in natural units}). \quad (20)$$

Viscosity: momentum transfer between adjacent simplices occurs through shared hinges. The maximum transfer rate through a hinge is limited by:

- $c = 2$: lattice propagation speed (derived from $(n_S, n_T) = (3, 2)$),
- 2π : one full $U(1)$ phase rotation (the period of the gauge connection).

The phase capacity (maximum gauge information throughput per hinge):

$$\Gamma = c \times 2\pi = 4\pi. \quad (21)$$

The shear viscosity is the absorption cross-section divided by the phase capacity:

$$\eta = \frac{A_h}{c \times 2\pi} = \frac{A_h}{4\pi}. \quad (22)$$

Dividing:

$$\boxed{\frac{\eta}{s} = \frac{A_h/(4\pi)}{A_h} = \frac{1}{4\pi} \approx 0.0796} \quad (23)$$

The hinge area A_h cancels. The result is a pure geometric constant, depending only on:

- $c = 2$ (from n_T/d_T over n_S/d_S , i.e., from $d = 5$),
- 2π (from $\pi_1(S^1) = \mathbb{Z}$, i.e., from $\mathbb{K} = \mathbb{C}$).

This is the Kovtun–Son–Starinets (KSS) bound, derived with zero free parameters and no AdS/CFT correspondence.

5.2 Comparison with AdS/CFT

	AdS/CFT	DRLT
Method	5D black hole dual	$\mathbb{C}^5$ simplex geometry
Extra dimension	Spatial (AdS_5)	Internal ($\mathbb{C}^5$)
Origin of 4π	Horizon area of 5D BH	$c \times 2\pi$ (lattice speed $\times$ U(1) period)
Free parameters	$\lambda \rightarrow \infty$ required	0
Applies to	$\mathcal{N} = 4$ SYM (not QCD)	Any deconfined $\mathbb{C}^5$ state
Result type	Bound ($\eta/s \geq$)	Exact value ($\eta/s =$)

Remark 5.1. The “5” in AdS_5 and the “5” in $\mathbb{C}^5$ are the same 5: $d = 5$ from the atomic uniqueness theorem. AdS/CFT works because it accidentally accesses the correct dimensionality of the internal space. The “holographic correspondence” is a shadow of the Gram matrix structure.

6 η/s Profile Inside a Neutron Star

The viscosity-to-entropy ratio varies with depth:

ρ/ρ_0	$\det(G_h)$	$\mathbb{C}^5$ active channels	η/s	Region
1	0.35	7/25	$\gg 1/(4\pi)$	Outer core (viscous fluid)
3	0.18	16/25	$\sim 3/(4\pi)$	Inner core (transition)
5	0.09	20/25	$\sim 1.5/(4\pi)$	Deep core (approaching)
7	0.05	23/25	$\approx 1/(4\pi)$	Quark core onset
10	0.02	24/25	$= 1/(4\pi)$	Perfect fluid core

Prediction: The transition from viscous nuclear fluid to perfect quark fluid occurs at $\rho \approx 7\rho_0$. This transition is continuous (crossover), not first-order, because $\det(G_h)$ varies smoothly.

7 Summary of Predictions

#	Prediction	Value	Verification
38	Pure quark stars are unstable	—	Absence of observation
39	Max neutron star density	$\sim 5\rho_0$	GW tidal deformability
40	Quark core threshold	$\gtrsim 2 M_\odot$	GW merger waveforms
41	Core η/s	$1/(4\pi)$	Heavy-ion data
42	Deconfinement crossover	$\rho \approx 7\rho_0$	NICER X-ray data
43	$P_{\max}$ exists	$\sim 5\rho_0$	EOS from GW
44	$\hbar_{\text{eff}}$ varies with density	$\hbar \propto \sqrt{\det}$	Precision spectroscopy

All predictions are derived from $d = 5$ with zero free parameters, zero equation-of-state assumptions, and no nuclear physics input beyond the DRLT Gram matrix.

8 Conclusion

The stability of compact stars in DRLT reduces to a single question: *is $\mathbb{C}^3$ confinement active?*

- **Yes** (neutron star): $\det(G_h)$ has a floor $\rightarrow \hbar_{\text{eff}} > 0 \rightarrow P_{\text{deg}} > 0 \rightarrow$ stable.
- **No** (quark star): $\det(G_h)$ floor removed $\rightarrow$ positive feedback $\rightarrow \det \rightarrow 0^+ \rightarrow$ collapse.
- **Partial** (hybrid star): neutron shell provides external floor $\rightarrow$ quark core stable inside shell.

The $\mathbb{C}^3$ boundary, which in particle physics manifests as color confinement, plays a second role in astrophysics: it is the **structural element that prevents gravitational collapse**. Confinement is not just a property of the strong force — it is the reason neutron stars exist.

Confinement saves stars.

References

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